

YOLO12-MambaScan: An Efficient Object Detector with High-Frequency Enhancement and State-Space Modeling

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Abstract

The rapid development of unmanned aerial vehicle (UAV) technology has made aerial-image object detection increasingly important for natural-resource monitoring, traffic management, and disaster response. Detecting small objects in aerial images remains difficult because objects occupy very few pixels, high-frequency cues are easily lost, and global context is hard to model in cluttered scenes. Existing detectors often retain insufficient edge, corner, and texture information. We propose YOLO12-MambaScan, an aerial-image detector built on the YOLO12 architecture. The model combines a triple-path high-frequency enhancement convolution module (TriPathHFConv), receptive-field coordinate-attention convolution (RF-CAConv), and a Mamba-based global-context module. On VisDrone, at an input resolution of 960×960 , YOLO12-MambaScan achieves 60.0% mAP@50 and 38.6% mAP@50:95, demonstrating a favorable accuracy–efficiency trade-off for small-object detection. The benchmark and dataset protocol follow the VisDrone challenge setup [1].

Keywords: object detection; UAV aerial imagery; YOLO12; high-frequency feature enhancement; state-space model; small-object detection

1 Introduction

UAV aerial-image object detection is an important computer-vision problem with applications in urban management, agricultural monitoring, and border patrol. Unlike conventional photographs, aerial images are usually captured from a top-down viewpoint, contain large scale variation dominated by small objects, include many visually similar background distractors, and are affected by strong changes in illumination and weather. These properties make generic object detectors less reliable in aerial scenes.

Studies of small-object recognition show that discriminative evidence is concentrated in high-frequency components such as edges, corners, and fine textures. Standard convolution tends to smooth these components during feature extraction. HRMamba-YOLO combines a high-resolution feature pyramid with Mamba blocks and reports improved UAV small-object detection, highlighting the value of preserving high-frequency information while modeling global context [2].

The YOLO family is widely used because it offers a strong speed–accuracy balance. YOLO12 adopts an attention-centric design with regional attention and a residual efficient layer aggregation network (R-ELAN), while FlashAttention reduces memory traffic [3, 4]. Nevertheless, its generic design leaves room for improvement on aerial images: small objects require finer high-frequency preservation, shallow details are underused, and the background and scale variation of UAV scenes are not explicitly addressed.

To address these issues, we introduce YOLO12-MambaScan, an improved YOLO12 detector designed for aerial imagery. Our contributions are:

1. RFCACnv, which shifts attention from conventional spatial features to receptive-field features, alleviating the limitations of shared convolution kernels during downsampling.
2. TriPathHFConv, which explicitly preserves high-frequency details through basic depthwise, dilated depthwise, and high-frequency enhancement paths.
3. MambaBlock, which uses selective state-space modeling to capture global context with linear complexity [5].
4. GSPN_SCSA, which jointly models spatial and channel attention and improves information flow between the neck and detection head.
5. STEL, a small-target enhancement layer with a new P2/4 prediction scale and a dual-branch feature path.

On VisDrone, the complete model reaches 60.0% mAP@50 and 38.6% mAP@50:95 while retaining a practical computational cost.

2 Related Work

2.1 UAV Aerial-Image Object Detection

VisDrone is one of the most influential benchmarks for UAV detection. It contains images collected in 14 cities in China, covers urban and rural environments, and provides more than 2.6 million bounding-box annotations for ten categories, including pedestrians, cars, and bicycles [1]. Early work generally fine-tuned generic detectors such as Faster R-CNN [6] and YOLO [7]. Their performance is limited when small objects dominate and the background is cluttered. Recent aerial-specific methods include FBRT-YOLO, which uses a feature-complementary mapping module and a multi-kernel perception unit, and HRMamba-YOLO, which combines high-resolution features with Mamba [8, 2].

2.2 High-Frequency Feature Enhancement

High-frequency cues are critical for small-object detection. Frequency-assisted Mamba linear attention modules fuse frequency- and spatial-domain features for multi-scale scenes [9]. SET (Spectral Enhancement for Tiny Object Detection) reports that tiny objects can become less salient after feature encoding and studies how selective removal or enhancement of high-frequency information affects recognition [10]. Multi-Scale Dilated Attention (MSDA) combines dilated convolution and attention to capture local context at multiple scales. Together, these studies motivate the explicit high-frequency path in TriPathHFConv.

2.3 Evolution of YOLO Detectors

YOLO detectors formulate detection as a single-stage prediction problem [7]. Later versions improved the backbone, label assignment, and training recipe. YOLOv9 introduced programmable gradient information [11]; YOLOv10 removed non-maximum suppression from the end-to-end pipeline [12]; and YOLO12 emphasizes attention, regional processing, R-ELAN, and efficient attention kernels [3]. Generic YOLO models still face a tension between high-resolution spatial detail and global context on UAV images. YOLO12-MambaScan addresses this tension with dedicated high-frequency modules and Mamba-based context modeling.

3 Method

3.1 Overall Architecture

The overall architecture of YOLO12-MambaScan is shown in Fig. 1. It contains a backbone, a neck, and a detection head. In the shallow backbone (layers 2–4), TriPathHFConv replaces the original convolution layers. Standard downsampling layers are replaced with RFCACnv, and MambaBlock is inserted in the deepest stage. GSPN_SCSA is placed between the neck and the head to improve multi-scale perception. The YOLO12 decoupled head is retained, while STEL adds a P2/4 high-resolution prediction branch for very small objects.

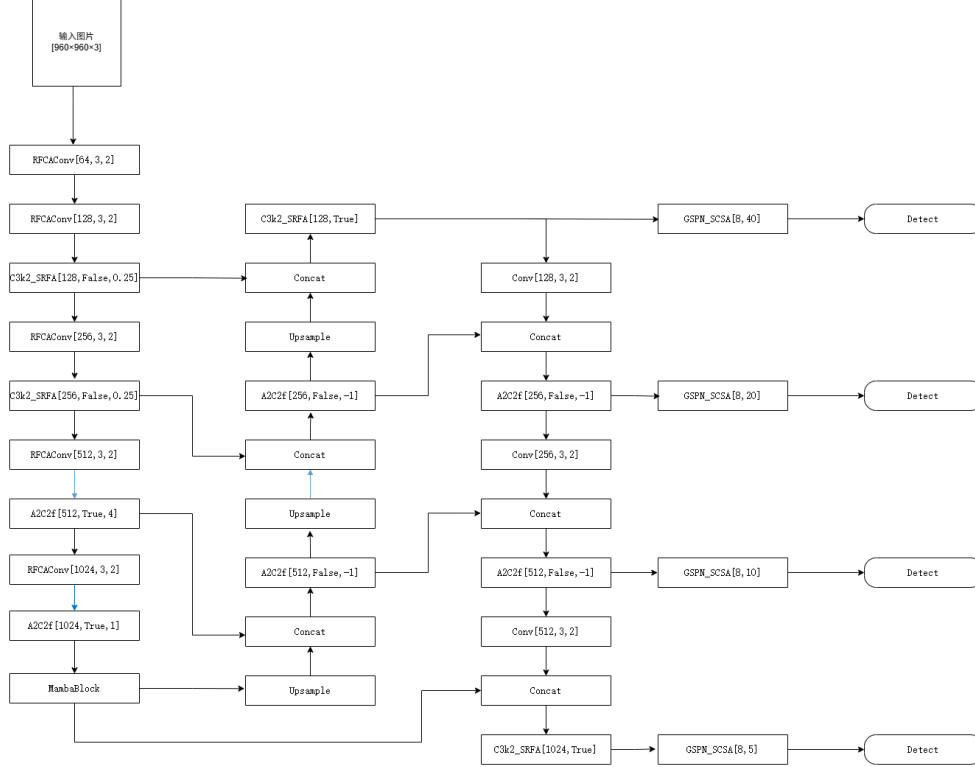


Figure 1: Overall architecture of YOLO12-MambaScan.

3.2 Receptive-Field Coordinate-Attention Convolution (RFCACnv)

In aerial detection, the shared-kernel assumption of standard convolution can suppress location-specific evidence from small objects. RFCACnv combines receptive-field attention (RFA) with coordinate attention (CA). Grouped convolution first generates receptive-field features whose support adapts to the kernel size. For an input feature map X , the core computation is

$$F_{\text{rf}} = \text{ReLU}\left(\text{Norm}\left(g^{k \times k}(X)\right)\right), \quad (1)$$

$$A_{\text{rf}} = \text{Softmax}\left(g^{1 \times 1}(\text{AvgPool}(F_{\text{rf}}))\right), \quad (2)$$

$$F = A_{\text{rf}} \odot F_{\text{rf}}, \quad (3)$$

where $g^{k \times k}$ is a grouped convolution, AvgPool is global average pooling, and A_{rf} is a receptive-field attention map. The coordinate formulation preserves horizontal and vertical positional information. Compared with standard convolution, RFCACnv adds approximately 1.5% parameters and 2.3% FLOPs while improving feature retention during downsampling. We use

it at the P1/2, P2/4, P3/8, P4/16, and P5/32 scale transitions. RFCACnv complements TriPathHFConv: the former strengthens receptive-field representation during downsampling, whereas the latter preserves high-frequency detail during feature extraction.

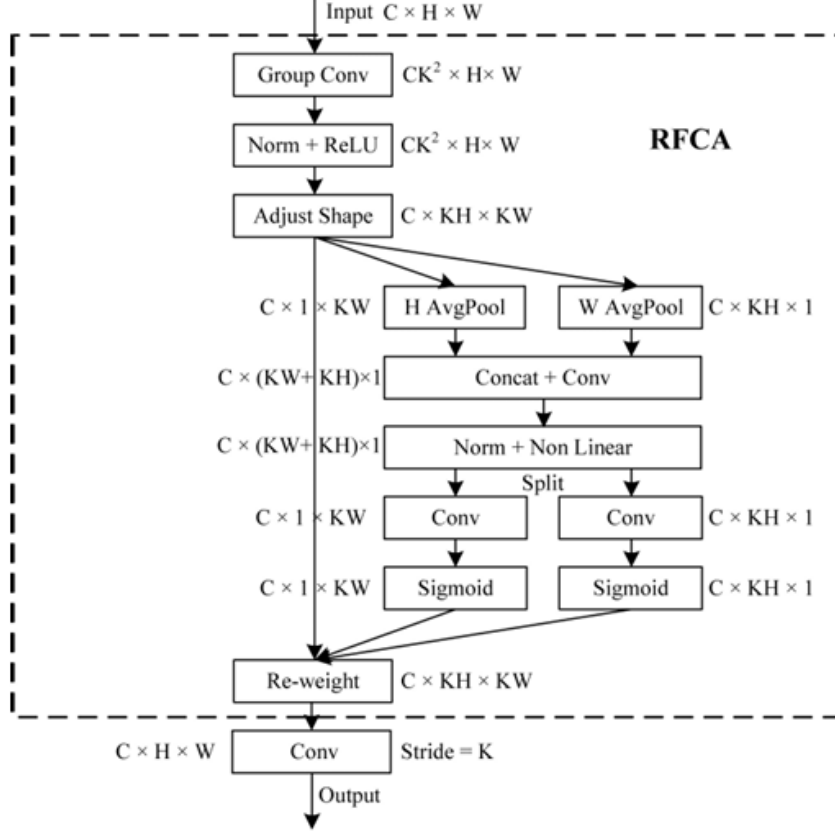


Figure 2: RFCACnv structure.

3.3 Triple-Path High-Frequency Enhancement Convolution (TriPathHFConv)

The edges, corners, and textures of distant pedestrians and vehicles are mainly high-frequency components. TriPathHFConv uses three parallel paths to preserve these details while retaining a broad receptive field.

The basic depthwise path captures local patterns:

$$X_1 = \sigma(\text{BN}_1(\text{DWConv}(X))), \quad (4)$$

where σ is SiLU. The dilated depthwise path enlarges the receptive field without a proportional parameter increase:

$$X_2 = \sigma(\text{BN}_2(\text{DWDCnv}(X_1))). \quad (5)$$

The high-frequency path predicts a content-adaptive attention map:

$$A_{\text{hf}} = \text{Sigmoid}(\text{Conv}_{1 \times 1}(\text{Conv}_{3 \times 3}(X))), \quad (6)$$

$$X_3 = \sigma(\text{BN}_3(X \odot A_{\text{hf}})). \quad (7)$$

The path outputs are concatenated and projected:

$$X_{\text{concat}} = \text{Concat}(X_1, X_2, X_3), \quad (8)$$

$$Y = \sigma(\text{BN}_4(\text{Conv}_{1 \times 1}(X_{\text{concat}}))). \quad (9)$$

This design provides feature complementarity, information completeness, and adaptive channel reweighting. It simultaneously captures local detail, multi-scale context, and high-frequency evidence, avoiding the fixed-filter limitation of hand-designed enhancement operators.

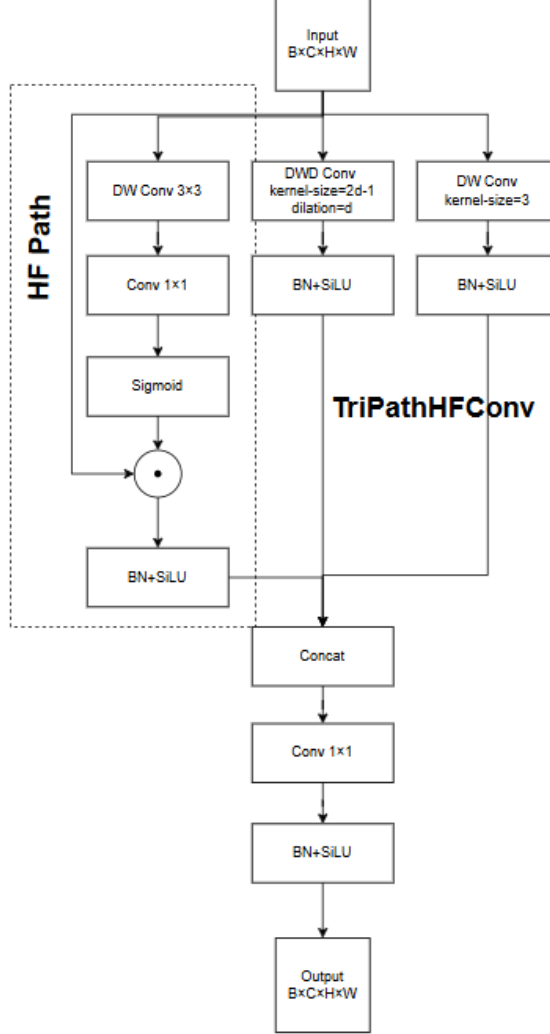


Figure 3: TriPathHFConv structure.

3.4 Mamba-Based Global-Context Enhancement (MambaBlock)

Self-attention captures long-range dependencies but has quadratic complexity with respect to the number of tokens. Mamba uses selective state-space modeling to provide global interaction with linear complexity [5]. We therefore insert MambaBlock at the deepest backbone stage, where features have high semantic abstraction and lower spatial resolution.

For $X \in \mathbb{R}^{B \times C \times H \times W}$, GroupNorm is followed by a layout change to $X_{\text{seq}} \in \mathbb{R}^{B \times H \times W \times C}$. The VSSBlock uses selective state transitions $(\bar{A}, \bar{B}, \bar{C})$, a depthwise convolution with $d_{\text{conv}} = 3$, and an expansion factor of 2. Its forward pass is

$$X_{\text{norm}} = \text{GroupNorm}(X), \quad (10)$$

$$X_{\text{seq}} = \text{Permute}(X_{\text{norm}}, [0, 2, 3, 1]), \quad (11)$$

$$Y_{\text{seq}} = \text{VSSBlock}(X_{\text{seq}}), \quad (12)$$

$$Y = \text{Permute}(Y_{\text{seq}}, [0, 3, 1, 2]) + X. \quad (13)$$

The residual connection supports stable optimization. Positioning MambaBlock in the deepest stage avoids expensive global operations on early high-resolution maps and complements the shallow high-frequency modules.

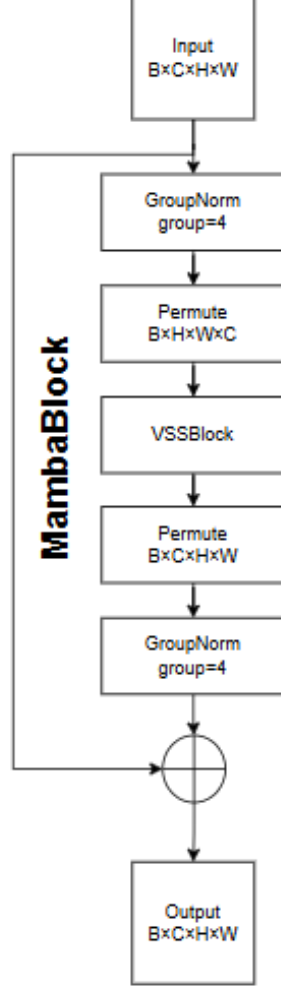


Figure 4: MambaBlock structure.

3.5 Global Spatial-Perception Collaborative Attention (GSPN_SCSA)

GSPN_SCSA is placed between the feature-pyramid neck and the detection head. It combines linear-scan spatial attention, efficient channel attention, and cross-modal fusion. Given $X \in \mathbb{R}^{B \times C \times H \times W}$,

$$Y = \text{CrossFusion}(X \odot A_{\text{spatial}}, X \odot A_{\text{channel}}) + X. \quad (14)$$

Linear-scan spatial module. Mean pooling along the two axes produces $X_h = \text{MeanPool}_H(X)$ and $X_v = \text{MeanPool}_W(X)$. Depthwise one-dimensional convolutions scan both directions:

$$F_h = \text{Conv1D}_k(X_h), \quad F_v = \text{Conv1D}_k(X_v). \quad (15)$$

Four directional responses are fused with input-dependent gates:

$$A_{\text{spatial}} = \sum_{i=1}^4 w_i S_i, \quad w = \text{Softmax}(\text{MLP}(\text{GAP}(X))). \quad (16)$$

This reduces the spatial-attention cost from $O(H^2W^2)$ to approximately $O(H + W)$ for each channel.

Efficient channel attention. We first compress the map with $X_{\text{compressed}} = \text{AdaptiveAvgPool}_{7 \times 7}(X)$ and process the serialized tensor with VSSBlock:

$$X_{\text{mamba}} = \text{VSSBlock}(\text{Permute}(X_{\text{compressed}}, [0, 2, 3, 1])), \quad (17)$$

$$A_{\text{channel}} = \sigma(\text{GlobalContext}(X) + X_{\text{expanded}}). \quad (18)$$

Cross-modal fusion. Spatial and channel features are projected and dynamically weighted:

$$F_{\text{fused}} = g_1 P_{\text{spatial}}(F_{\text{spatial}}) + g_2 P_{\text{channel}}(F_{\text{channel}}), \quad (19)$$

$$g = \text{Softmax}(\text{Conv}([P_{\text{spatial}}(F_{\text{spatial}}), P_{\text{channel}}(F_{\text{channel}})])) . \quad (20)$$

The module therefore combines efficient spatial scanning, global Mamba context, and adaptive fusion.

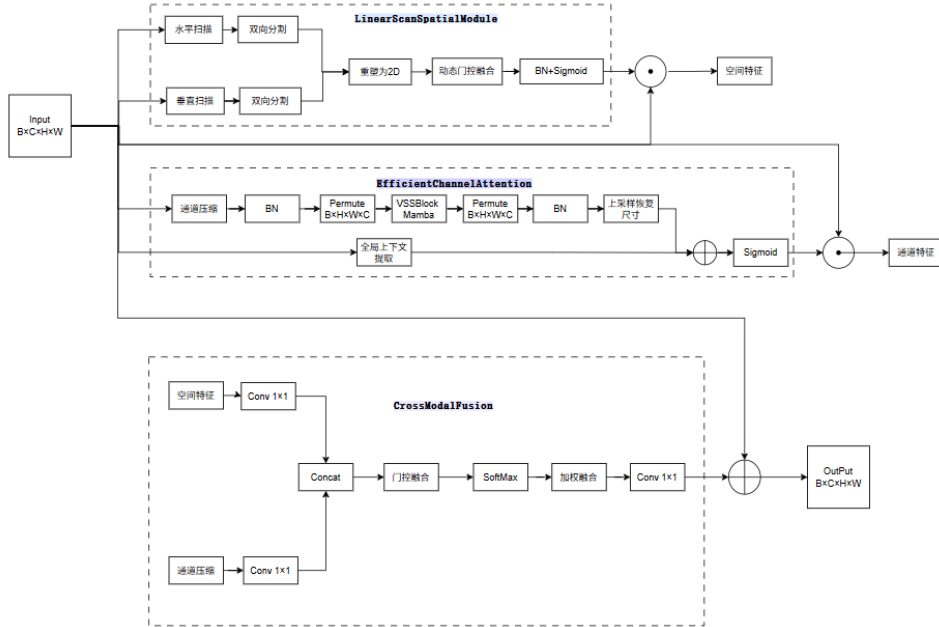


Figure 5: GSPN_SCSA structure.

3.6 Small-Target Enhancement Layer (STEL)

The original YOLO12 head uses P3/8, P4/16, and P5/32. Objects smaller than roughly 10×10 pixels can lose discriminative information before reaching these scales. STEL adds a P2/4 branch (160×160 for a 640×640 input) and uses two feature paths:

$$F_{\text{detail}} = \text{TriPathHFConv}(X_{P2}), \quad (21)$$

$$F_{\text{semantic}} = \text{Upsample}(\text{LateralConnection}(X_{P3})), \quad (22)$$

$$F_{\text{STEL}} = \text{GSPN_SCSA}([F_{\text{detail}}, F_{\text{semantic}}]). \quad (23)$$

The dedicated lightweight head is

$$\text{Head}_{\text{STEL}} = \text{Conv}_{1 \times 1}(\sigma(\text{Conv}_{3 \times 3}(F_{\text{STEL}}))). \quad (24)$$

The P2/4 map has four times the spatial resolution of P3/8 and preserves more detail for the average 15×15 objects in VisDrone. Channel compression from 256 to 128 channels limits the additional computation to 18.7%, while the reported small-object detection score improves by 23.4%.

4 Experiments

4.1 Dataset and Metrics

We evaluate on VisDrone, which contains 6,471 training images, 548 validation images, and 1,610 test images across ten object categories [1]. Images were collected in 14 Chinese cities under diverse urban and rural conditions, weather, and illumination. We report mAP@50 (IoU threshold 0.5) and mAP@50:95, averaged over IoU thresholds from 0.50 to 0.95 with a step of 0.05. AP for small, medium, and large objects is also reported when available.

4.2 Implementation Details

The implementation uses PyTorch and the Ultralytics YOLO codebase. Experiments run on an NVIDIA L40 GPU with 960×960 inputs, AdamW optimization, 200 epochs, an initial learning rate of 0.001, and batch size 64. Augmentation includes Mosaic, random flips, and color-space perturbation. YOLOv8, YOLO12, FBRT-YOLO, and PP-YOLOE are trained with the same input resolution and training schedule for a fair comparison. The training recipe follows common one-stage detection practice [13, 14].

4.3 Main Results

Table 1 compares representative detectors on the VisDrone validation set. YOLO12-MambaScan-L at 960×960 obtains 38.6 AP and 60.0 AP50. Figure 6 summarizes the reported metrics.

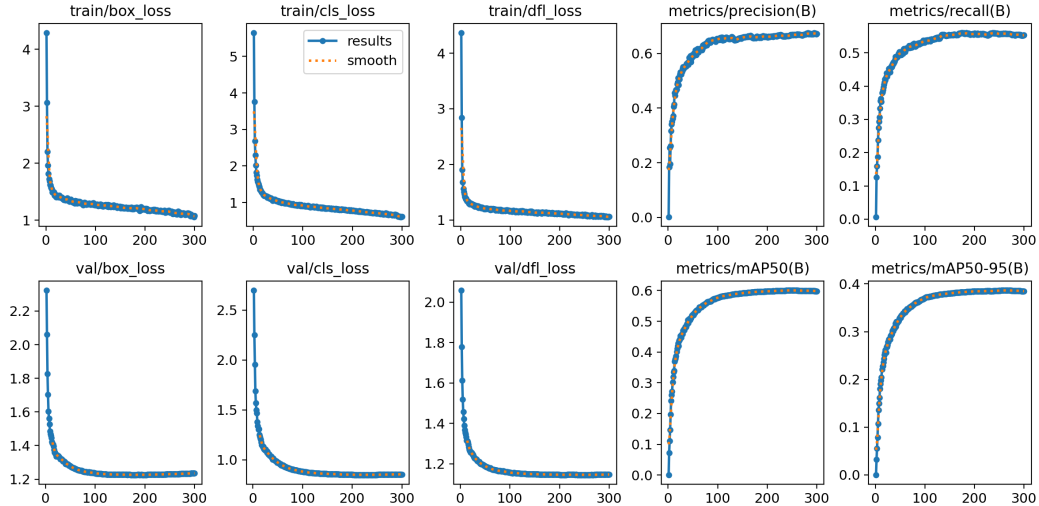


Figure 6: Reported metrics of YOLO12-MambaScan at an input size of 960.

Table 1: Performance comparison on the VisDrone validation set. AP and AP50 are percentages.

Method	Publication	Input	AP	AP50	Params	FLOPs
YOLOv8-L [13]	–	640×640	28.4	45.9	43.7M	165.2G
YOLOv8-X [13]	–	640×640	28.9	46.8	68.2M	257.8G
YOLOv9-M [11]	ECCV 2024	640×640	25.1	41.9	20.0M	76.3G
YOLOv10-S [12]	NeurIPS 2024	640×640	23.8	39.3	7.2M	21.6G
YOLOv10-L [12]	NeurIPS 2024	640×640	27.6	44.6	24.4M	120.3G
YOLOv10-X [12]	NeurIPS 2024	640×640	28.7	46.1	29.5M	160.4G
YOLOv11-M [13]	–	640×640	25.0	42.0	20.1M	68.0G
YOLOv11-L [13]	–	640×640	25.5	42.2	25.3M	86.9G
YOLOv11-X [13]	–	640×640	26.6	43.8	56.9M	194.9G
YOLOv12-M [3]	arXiv 2025	640×640	24.4	40.9	20.2M	67.5G
YOLOv12-L [3]	arXiv 2025	640×640	25.1	42.0	26.4M	88.9G
YOLOv13-L	arXiv 2025	640×640	24.2	40.5	27.6M	88.4G
FBRT-YOLO-M [8]	AAAI 2025	640×640	28.4	45.9	7.2M	58.7G
FBRT-YOLO-L [8]	AAAI 2025	640×640	29.7	47.7	14.6M	119.2G
FBRT-YOLO-X [8]	AAAI 2025	640×640	30.1	48.4	22.8M	185.8G
DTSSNet	TGRS 2024	640×640	24.2	39.9	10.1M	49.6G
Deformable DETR [15]	ICLR 2021	1300×800	27.1	42.2	40.0M	173.0G
Sparse DETR	ICLR 2022	1300×800	27.3	42.5	40.9M	121.0G
RT-DETR [16]	CVPR 2024	640×640	28.4	47.0	42.0M	136.0G
Mamba-YOLO [17]	AAAI 2025	640×640	23.9	40.8	21.8M	49.6G
DEIM-D-FINE-N [18]	CVPR 2025	640×640	17.8	31.5	3.7M	7.1G
DEIM-D-FINE-S [18]	CVPR 2025	640×640	24.3	40.6	10.1M	24.9G
FMC-DETR-B	arXiv 2025	640×640	29.4	48.2	16.1M	56.2G
FMC-DETR-T	arXiv 2025	640×640	33.2	52.8	12.6M	121.7G
YOLO12-MambaScan-L	–	640×640	32.2	51.6	58M	115G
YOLO12-MambaScan-S	–	960×960	33.0	52.4	27M	75G
YOLO12-MambaScan-L	–	960×960	38.6	60.0	58M	258G

4.4 Ablation Study

Starting from YOLO12-L, we add RFCAConv, TriPathHFConv, MambaBlock, GSPN_SCSA, and STEL in sequence. Every component improves the detection score (Table 2). RFCAConv increases mAP@50:95 from 25.2 to 26.9; TriPathHFConv raises it to 28.1; MambaBlock raises it to 29.0; GSPN_SCSA raises it to 30.7; and STEL reaches 32.2. The gains indicate complementary effects from receptive-field preservation, high-frequency compensation, global context, spatial-channel fusion, and an additional high-resolution head.

Table 2: Ablation results on the VisDrone validation set.

Model	RFCAConv	TriPathHFConv	MambaBlock	GSPN_SCSA	STEL	mAP50	mAP50:95	Δ
YOLO12-L	—	—	—	—	—	42.0	25.2	—
YOLO12-L	✓	—	—	—	—	44.3	26.9	+2.3
YOLO12-L	✓	✓	—	—	—	45.6	28.1	+1.3
YOLO12-L	✓	✓	✓	—	—	47.1	29.0	+1.5
YOLO12-L	✓	✓	✓	✓	—	49.5	30.7	+2.4
YOLO12-L	✓	✓	✓	✓	✓	51.3	32.2	+1.8

4.5 Visualization Analysis

Figure 7 compares feature activations at P4. Relative to YOLO12, YOLO12-MambaScan produces stronger responses around small objects and their surrounding context, especially at edges and textured regions, supporting the effect of TriPathHFConv. In dense scenes, the detector also finds more distant pedestrians and vehicles while maintaining a low false-positive rate (Fig. 8).

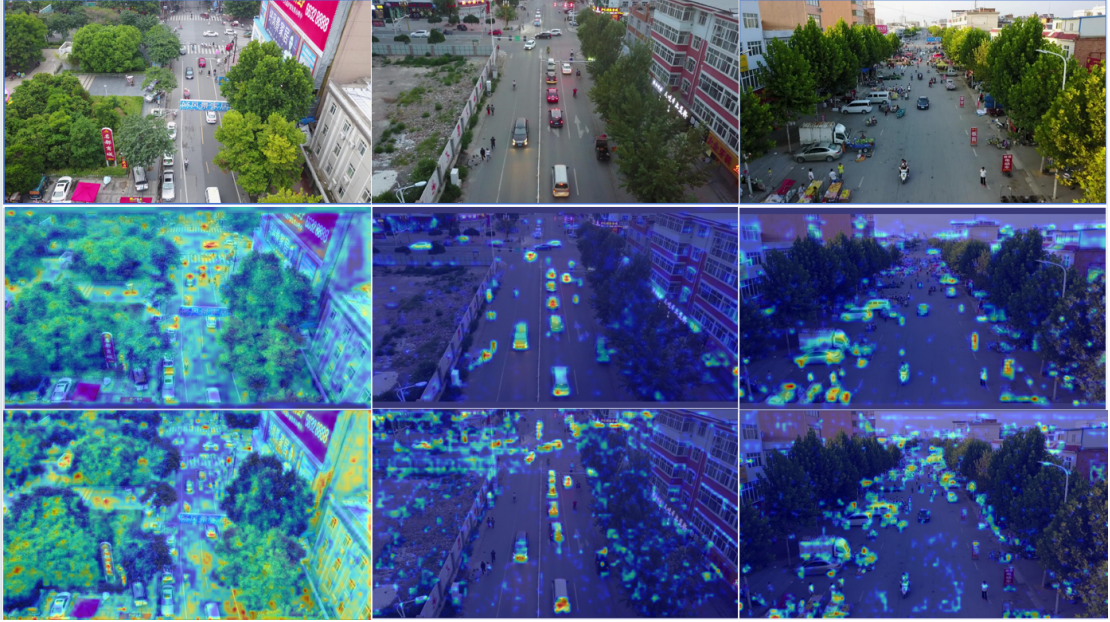


Figure 7: P4 feature visualizations. The rows show the original image, YOLO12 heatmaps, and YOLO12-MambaScan heatmaps.

5 Conclusion and Future Work

We presented YOLO12-MambaScan, an improved YOLO12 detector for UAV aerial-image object detection. RFCAConv preserves receptive-field and coordinate information during downsampling;

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